HOT WET EQUATORIAL CLIMATE

EQUATORIAL CLIMATE MAP → CONVECTION DAY CYCLE → TEMPERATURE-RANGE FIREWALL → RAINFALL IDENTITY → RAINFOREST VERTICAL PROFILE → NUTRIENT-CYCLE LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 EQUATORIAL CLIMATE MAP

EQUATORIAL CLIMATE MAP · PERSISTENT HIGH SOLAR RECEIPT · AMAZON + CONGO · MAJOR CONTINENTAL LOWLANDS · MALAYSIA + EAST INDIES · MARITIME EXPRESSION · ALTITUDE OR MONSOON MODIFICATION · MUST REMEMBER

EQUATOR → persistent high solar receipt

AMAZON + CONGO → major continental lowlands → MALAYSIA + EAST INDIES → maritime expression → HIGHLANDS / MARGINS → altitude or monsoon modification

CORE CLAIM
- EQUATOR → persistent high solar receipt
- AMAZON + CONGO → major continental lowlands

MECHANISM
- MALAYSIA + EAST INDIES → maritime expression
- HIGHLANDS / MARGINS → altitude or monsoon modification

CONSEQUENCE / LIMIT
- MUST REMEMBER: Equatorial Af climate follows year-round high sun, ITCZ convection, humid air and weak annual thermal range,
- producing frequent rainfall, evergreen stratification, intense chemical weathering and leached soils in Amazon, Congo and Maritime Southeast Asia.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Equatorial Af climate follows year-round high sun, ITCZ convection, humid air and weak annual thermal range, producing frequent rainfall, evergreen stratification, intense chemical weathering and leached soils in Amazon, Congo and Maritime Southeast Asia.

01

STAGE 01 CONVECTION DAY CYCLE

CONVECTION DAY CYCLE · MORNING SUN · MOIST SURFACE HEATING · NOON ASCENT · EXPANDING AND COOLING AIR · CUMULONIMBUS THUNDERSTORM · RAIN-COOLED AIR; CYCLE CAN RECUR

MORNING SUN → moist surface heating → NOON ASCENT → expanding and cooling air → AFTERNOON → cumulonimbus thunderstorm → EVENING → rain-cooled air; cycle can recur

CORE CLAIM
- MORNING SUN → moist surface heating
- NOON ASCENT → expanding and cooling air

MECHANISM
- AFTERNOON → cumulonimbus thunderstorm
- EVENING → rain-cooled air; cycle can recur

CONSEQUENCE / LIMIT
- MORNING SUN → moist surface heating
- NOON ASCENT → expanding and cooling air

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AFTERNOON → cumulonimbus thunderstorm; EVENING → rain-cooled air; cycle can recur.

02

STAGE 02 TEMPERATURE-RANGE FIREWALL

TEMPERATURE-RANGE FIREWALL · ANNUAL RANGE · USUALLY VERY SMALL · DIURNAL RANGE · LARGER THAN ANNUAL RANGE · WEAK SEASONALITY NEAR EQUATOR · MEANS DO NOT CAP DAILY EXTREMES

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ANNUAL RANGE → usually very small	CAUSE → weak seasonality near Equator	ANNUAL RANGE → usually very small
DIURNAL RANGE → larger than annual range	CAUTION → means do not cap daily extremes	DIURNAL RANGE → larger than annual range

CORE CLAIM
- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

MECHANISM
- CAUSE → weak seasonality near Equator
- CAUTION → means do not cap daily extremes

CONSEQUENCE / LIMIT
- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → means do not cap daily extremes.

03

STAGE 03 RAINFALL IDENTITY

RAINFALL IDENTITY · RAIN ALL MONTHS · SUPPORTS AF READING · TRUE DRY SEASON · TEST AM OR AW INSTEAD · SOURCE NOTES OFTEN 150-250 CM · DISTRIBUTION OUTRANKS ONE ANNUAL NUMBER

RAIN ALL MONTHS → supports Af reading

- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

- CAUSE → weak seasonality near Equator
- CAUTION → means do not cap daily extremes

- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → means do not cap daily extremes.

03

STAGE 03

RAINFALL IDENTITY

RAINFALL IDENTITY · RAIN ALL MONTHS · SUPPORTS AF READING · TRUE DRY SEASON · TEST AM OR AW INSTEAD · SOURCE NOTES OFTEN 150-250 CM · DISTRIBUTION OUTRANKS ONE ANNUAL NUMBER

RAIN ALL MONTHS → supports Af reading

TRUE DRY SEASON → test Am or Aw instead → TOTAL → source notes often 150-250 cm → RULE → distribution outranks one annual number

CORE CLAIM
- RAIN ALL MONTHS → supports Af reading
- TRUE DRY SEASON → test Am or Aw instead

MECHANISM
- TOTAL → source notes often 150-250 cm
- RULE → distribution outranks one annual number

CONSEQUENCE / LIMIT
- RAIN ALL MONTHS → supports Af reading
- TRUE DRY SEASON → test Am or Aw instead

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRUE DRY SEASON → test Am or Aw instead.

04

STAGE 04

RAINFOREST VERTICAL PROFILE

RAINFOREST VERTICAL PROFILE · CROWNS ABOVE MAIN ROOF · DENSE LIGHT-CAPTURING LAYER · SHADE-TOLERANT GROWTH · DARK, HUMID, RAPID DECAY

EMERGENTS → crowns above main roof

CANOPY → dense light-capturing layer → UNDERSTOREY → shade-tolerant growth → FLOOR → dark, humid, rapid decay

CORE CLAIM
- EMERGENTS → crowns above main roof
- CANOPY → dense light-capturing layer

MECHANISM
- UNDERSTOREY → shade-tolerant growth
- FLOOR → dark, humid, rapid decay

CONSEQUENCE / LIMIT
- EMERGENTS → crowns above main roof
- CANOPY → dense light-capturing layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UNDERSTOREY → shade-tolerant growth; FLOOR → dark, humid, rapid decay.

05

STAGE 05

NUTRIENT-CYCLE LOOP

NUTRIENT-CYCLE LOOP · RAPID WARM-WET DECOMPOSITION · ROOT MAT · RAPID NUTRIENT UPTAKE · PRINCIPAL NUTRIENT CAPITAL · HEAVY RAIN · LEACHING AFTER CLEARANCE

LITTER → rapid warm-wet decomposition → ROOT MAT → rapid nutrient uptake → BIOMASS → principal nutrient capital → HEAVY RAIN → leaching after clearance

CORE CLAIM
- LITTER → rapid warm-wet decomposition
- ROOT MAT → rapid nutrient uptake

MECHANISM
- BIOMASS → principal nutrient capital
- HEAVY RAIN → leaching after clearance

CONSEQUENCE / LIMIT
- LITTER → rapid warm-wet decomposition
- ROOT MAT → rapid nutrient uptake

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BIOMASS → principal nutrient capital; HEAVY RAIN → leaching after clearance.

06

STAGE 06

FALLOW DECISION

FALLOW DECISION · SMALL CLEARING + LONG FALLOW · SHORTENED FALLOW · DECLINING BIOMASS · STEEP SLOPE + REPEATED FIRE · EROSION RISK · TENURE + LIVELIHOOD · MANAGEMENT OUTCOME

SMALL CLEARING + LONG FALLOW → recovery

SHORTENED FALLOW → declining biomass → STEEP SLOPE + REPEATED FIRE → erosion risk → TENURE + LIVELIHOOD → management outcome

CORE CLAIM
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

MECHANISM
- STEEP SLOPE + REPEATED FIRE → erosion risk
- TENURE + LIVELIHOOD → management outcome

CONSEQUENCE / LIMIT
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

06

STAGE 06

FALLOW DECISION

FALLOW DECISION · SMALL CLEARING + LONG FALLOW · SHORTENED FALLOW · DECLINING BIOMASS · STEEP SLOPE + REPEATED FIRE · EROSION RISK · TENURE + LIVELIHOOD · MANAGEMENT OUTCOME

SMALL CLEARING + LONG FALLOW → recovery

SHORTENED FALLOW → declining biomass → STEEP SLOPE + REPEATED FIRE → erosion risk → TENURE + LIVELIHOOD → management outcome

CORE CLAIM
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

MECHANISM
- STEEP SLOPE + REPEATED FIRE → erosion risk
- TENURE + LIVELIHOOD → management outcome

CONSEQUENCE / LIMIT
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STEEP SLOPE + REPEATED FIRE → erosion risk.

07

STAGE 07

HUMID-TROPICS ECONOMY

HUMID-TROPICS ECONOMY · CAPITAL + LABOUR + EXPORT CROP · MIXED FOREST · SELECTIVE LOGGING DIFFICULTY · ROAD ACCESS · EXTRACTION GEOGRAPHY · CLIMATE ENABLES; INSTITUTIONS DECIDE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
PLANTATION → capital + labour + export crop	RIVER / ROAD ACCESS → extraction geography	PLANTATION → capital + labour + export crop
MIXED FOREST → selective logging difficulty	RULE → climate enables; institutions decide	MIXED FOREST → selective logging difficulty

CORE CLAIM
- PLANTATION → capital + labour + export crop
- MIXED FOREST → selective logging difficulty

MECHANISM
- RIVER / ROAD ACCESS → extraction geography
- RULE → climate enables; institutions decide

CONSEQUENCE / LIMIT
- PLANTATION → capital + labour + export crop
- MIXED FOREST → selective logging difficulty

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLANTATION → capital + labour + export crop.

08

STAGE 08

INDIA ANALOGUE MAP

INDIA ANALOGUE MAP · WESTERN GHATS · WINDWARD EVERGREEN BELT · WET EVERGREEN · EASTERN HIMALAYAN FOOTHILLS · INSULAR RAINFOREST

WESTERN GHATS → windward evergreen belt

NORTH-EAST → wet evergreen / semi-evergreen → EASTERN HIMALAYAN FOOTHILLS → transition → ANDAMAN-NICOBAR → insular rainforest

CORE CLAIM
- WESTERN GHATS → windward evergreen belt
- NORTH-EAST → wet evergreen / semi-evergreen

MECHANISM
- EASTERN HIMALAYAN FOOTHILLS → transition
- ANDAMAN-NICOBAR → insular rainforest

CONSEQUENCE / LIMIT
- WESTERN GHATS → windward evergreen belt
- NORTH-EAST → wet evergreen / semi-evergreen

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EASTERN HIMALAYAN FOOTHILLS → transition; ANDAMAN-NICOBAR → insular rainforest.

09

STAGE 09

OFFICIAL EVIDENCE FIREWALL

OFFICIAL EVIDENCE FIREWALL · FSI REPORT · DATED BIENNIAL ASSESSMENT · FOREST COVER · CANOPY-BASED MAPPED CLASS · FOREST TYPE · ECOLOGICAL CLASSIFICATION · LEGAL STATUS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FSI REPORT → dated biennial assessment	FOREST TYPE → ecological classification	CLOSE DISTINCTION: Equatorial does not mean everywhere exactly on the Equator, convectional rainfall is not continuous all-day
FOREST COVER → canopy-based mapped class	LEGAL STATUS → separate administrative category	rain, and luxuriant biomass does not imply nutrient-rich soil because rapid cycling stores nutrients mainly in biomass.

CORE CLAIM
- FSI REPORT → dated biennial assessment
- FOREST COVER → canopy-based mapped class

MECHANISM
- FOREST TYPE → ecological classification
- LEGAL STATUS → separate administrative category

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Equatorial does not mean everywhere exactly on the Equator, convectional rainfall is not continuous all-day
- rain, and luxuriant biomass does not imply nutrient-rich soil because rapid cycling stores nutrients mainly in biomass.

OFFICIAL EVIDENCE FIREWALL | FSI REPORT | DATED BIENNIAL ASSESSMENT | FOREST COVER | CANOPY-BASED MAPPED CLASS | FOREST TYPE | ECOLOGICAL CLASSIFICATION | LEGAL STATUS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FSI REPORT → dated biennial assessment	FOREST TYPE → ecological classification	CLOSE DISTINCTION: Equatorial does not mean everywhere exactly on the Equator, convectional rainfall is not continuous all-day
FOREST COVER → canopy-based mapped class	LEGAL STATUS → separate administrative category	rain, and luxuriant biomass does not imply nutrient-rich soil because rapid cycling stores nutrients mainly in biomass.

CORE CLAIM

- FSI REPORT → dated biennial assessment
- FOREST COVER → canopy-based mapped class

MECHANISM

- FOREST TYPE → ecological classification
- LEGAL STATUS → separate administrative category

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Equatorial does not mean everywhere exactly on the Equator, convectional rainfall is not continuous all-day
- rain, and luxuriant biomass does not imply nutrient-rich soil because rapid cycling stores nutrients mainly in biomass.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Equatorial does not mean everywhere exactly on the Equator, convectional rainfall is not continuous all-day rain, and luxuriant biomass does not imply nutrient-rich soil because rapid cycling stores nutrients mainly in biomass.

10

STAGE 10 VERIFIED PYQ PAIR

VERIFIED PYQ PAIR | RAINFOREST VEGETATION STRUCTURE | NUTRIENTS AND DECOMPOSITION | OFFICIAL ANSWER LETTERS WITHHELD | TEACH DISTINCTION; DO NOT INVENT KEY | EVIDENCE LIMIT | DEFORESTATION ALTERS ENERGY, MOISTURE AND NUTRIENT CYCLES BUT RAINFALL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
2021 → rainforest vegetation structure	LEDGER → official answer letters withheld	EVIDENCE LIMIT: Deforestation alters energy, moisture and nutrient cycles but rainfall feedback magnitude varies by scale and region
2023 → nutrients and decomposition	METHOD → teach distinction; do not invent key	Indian evergreen forests are analogues with monsoonal/topographic differences, not Af replicas.

CORE CLAIM

- 2021 → rainforest vegetation structure
- 2023 → nutrients and decomposition

MECHANISM

- LEDGER → official answer letters withheld
- METHOD → teach distinction; do not invent key

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Deforestation alters energy, moisture and nutrient cycles but rainfall feedback magnitude varies by scale and region
- Indian evergreen forests are analogues with monsoonal/topographic differences, not Af replicas.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Deforestation alters energy, moisture and nutrient cycles but rainfall feedback magnitude varies by scale and region; Indian evergreen forests are analogues with monsoonal/topographic differences, not Af replicas.

11

STAGE 11 EQUATORIAL ANSWER SPINE

EQUATORIAL ANSWER SPINE | EQUATORIAL LOWLAND BELT | HEAT, CONVERGENCE, CONVECTION, RAIN | LAYERS, NUTRIENTS AND LAND USE | INDIA ANALOGUE AND SOURCE LIMITS

LOCATE → equatorial lowland belt → DERIVE → heat, convergence, convection, rain → LINK → layers, nutrients and land use → QUALIFY → India analogue and source limits

CORE CLAIM

- LOCATE → equatorial lowland belt
- DERIVE → heat, convergence, convection, rain

MECHANISM

- LINK → layers, nutrients and land use
- QUALIFY → India analogue and source limits

CONSEQUENCE / LIMIT

- LOCATE → equatorial lowland belt
- DERIVE → heat, convergence, convection, rain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK → layers, nutrients and land use; QUALIFY → India analogue and source limits.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | DO NOT GENERALISE | MOST INDIAN EVERGREEN BELTS ARE MONSOONAL, WHILE LIMITED AF | INDIA'S CLOSEST EQUIVALENT IS TROPICAL WET EVERGREEN FOREST IN | MAIN REGIONS | WINDWARD WESTERN GHATS | MALABAR , NORTH-EAST INDIA , ANDAMAN & NICOBAR ISLANDS | MALABAR VEGETATION RANGES FROM MOIST TROPICAL EVERGREEN TO MIXED

OPTIONAL SOURCE DEPTH

- Do not generalise: most Indian evergreen belts are monsoonal, while limited Af classification depends on the map/dataset.
- India's closest equivalent is tropical wet evergreen forest in very high-rainfall belts.
- Main regions: windward Western Ghats / Malabar , North-East India , Andaman & Nicobar Islands .
- Malabar vegetation ranges from moist tropical evergreen to mixed broad-leaved and monsoon deciduous types.

ADVANCED DEBATE

- Andaman-Nicobar has tropical evergreen/semi-evergreen and littoral forests with high endemism.
- The archipelago combines forest and reef biodiversity with high seismicity and Barren Island volcanism.
- Western Ghats + North-East India are biodiversity-hotspot landscapes.
- Jhum is shifting cultivation; degradation rises when fallow cycles shorten.

LEGACY / NUANCE

- NOT: India has a true equatorial Af climate in the south → India lacks true Af; it has equatorial-type evergreen forests in wet monsoon belts.
- NOT: The leeward Deccan side of the Western Ghats has evergreen rainforest → Evergreen forest is on the windward western slope
- the leeward side is drier.
- NOT: Andaman forests are ordinary dry deciduous forests → A&N contains tropical evergreen,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels